PRODUCT SPECIFICATION

amplifi Quotes: Product Specification

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01 Summary

amplifi Quotes is a quoting tool for companies that sell services with real pricing math: labor by role, travel, lodging, per diems, tiered plans, regional rates, equipment-based pricing, parts, and recurring contracts. A sales rep builds a quote in a guided flow. Pricing rules live in settings, not code. The customer gets a branded link where they can pick an option, accept, sign, and download a PDF.

It ships two ways from one codebase:

- **Free, self-deployed.** A Deploy to Cloudflare button puts a full copy into the customer's own Cloudflare account on Cloudflare's free plan. The customer owns the data. amplifi runs nothing for them.
- **Paid, hosted by amplifi.** The same app runs as a multi-workspace service on amplifi's Cloudflare account, with custom domains, billing, and support.

The prototype is a single-company quoting system amplifi built and runs in production. It proved the workflow: 7 quote types, regional rates, approvals, revisions, a customer quote page, PDFs, and 7 languages. The prototype hard-codes its pricing (146 branches on quote type, 188 references to its tier names), so this product is a new build that carries over the prototype's behavior and its tested arithmetic, not its code structure.

02 Goals

ID	GOAL	MEASURE
G-1	A non-developer deploys a working copy in one click	Fresh Cloudflare account to first saved quote in under 10 minutes, no terminal
G-2	Pricing is configuration	Every prototype quote type rebuilt as a template with zero custom code
G-3	Totals are always right and always explainable	Every total has a line-by-line trace. Pricing checks block any change that moves a saved example without sign-off
G-4	Runs inside Cloudflare's free plan for a typical team	50 people quoting all day fits the free daily limits with room to spare
G-5	One codebase for both editions	Self-deployed and hosted builds pass the same test suite

03 Principles

1. **The server decides the price.** The browser shows a live preview, the Worker recomputes on every save, and the stored result always comes from the server.
2. **Nothing changes a sent quote.** Sent revisions are frozen: prices, exchange rates, terms, and the template version.

3. **Pricing changes are tested before they publish.** Saved examples act as tests. A change that moves one needs a person to accept the new number.
4. **Customer data stays with the customer.** The self-deployed edition sends nothing to amplifi. No telemetry, no phone-home, update checks off unless the owner turns them on.
5. **Free plan first.** Every feature has a free-plan budget. Features that can't fit are optional and say so on screen.
6. **No passwords.** Sign-in uses passkeys and email links. Password hashing alone (57 to 70 ms per login, measured) breaks the free plan's 10 ms compute limit, and reused passwords are the most common breach path.
7. **Plain screens.** Labels, not explainers. No decorative animation.

04 Out of scope

ITEM	REASON
Invoicing, payment collection after acceptance, accounting	Hand off to Stripe, QuickBooks, or the CRM. Phase 5 adds a deposit at acceptance only
Work orders, scheduling, dispatch	Field service platforms own this. Webhooks let them pick up accepted quotes
Contract lifecycle management (redlines, clause libraries)	Terms are a template per workspace. Negotiated contracts stay in the customer's legal tools
Inventory and stock levels	Catalog holds prices, not quantities on hand
Native mobile apps	The web app is phone-ready. Reps and customers both work in the browser
Self-hosting outside Cloudflare (Docker, on-prem)	Considered after launch. The adapter layer keeps it possible
Server-side headless Chrome	Cloudflare's free plan allows 10 minutes a day. PDFs and preview images render in the browser instead

05 Roles

Each workspace has members with one role. The customer is not a member; they reach a quote through its link.

ROLE	CAN	CAN'T
Owner	Everything, including billing (hosted), deleting the workspace, restoring backups	Nothing restricted
Admin	Members, roles, settings, branding, integrations, API keys, all quotes	Billing, delete workspace, restore backups
Pricing editor	Catalog, rates, regions, templates, terms, pricing checks; publish pricing changes	Members, integrations
Approver	Approve or reject quotes over a threshold; view all quotes; everything Sales can do	Pricing, settings
Sales	Create, edit, send, and revise their own quotes; view team quotes when the workspace allows it	Approve own quotes over threshold, pricing
Viewer	Read all quotes and reports	Any change
Customer (external)	Open the quote link, pick options, accept, sign, comment, download the PDF	See internal notes, cost, margin, traces, or other quotes

Appendix A lists every action by role.

06 Features

Each requirement has an ID, a test for done, and a phase (P0 to P5, defined in section 13).

6.1 One-click deploy and first-run setup (DEP)

ID	REQUIREMENT	DONE WHEN	PHASE
DEP-1	The public repo README and the amplifi site carry a Deploy to Cloudflare button pointing at the repo root	Clicking it opens Cloudflare's deploy page with the project prefilled	P0
DEP-2	The deploy creates every resource the app needs from `wrangler.jsonc`: one Worker with static assets and one SQLite-backed Durable Object class. No D1, KV, R2, or Queues in the default deploy	A fresh free account deploys with no manual resource setup and no payment method	P0
DEP-3	The setup page asks for one secret, `SETUP_CODE`, with help text from `package.json` `cloudflare.bindings`	The deploy page shows the field and its description	P0
DEP-4	First visit to the deployed URL shows a setup screen. Entering the setup code lets that person create the Owner account with a passkey	Wrong code is refused and rate limited (5 tries per 10 minutes per IP); right code creates the Owner once, then the setup screen never shows again	P0
DEP-5	Until setup finishes, every route except the setup screen and static assets returns 503	A scripted request to `/api/quotes` before setup gets 503	P0
DEP-6	Setup wizard: company name, logo, base currency, default language, time zone, first region, and a starter template to install	Owner reaches an empty dashboard with one template installed in under 3 minutes	P1
DEP-7	Setup prints 10 one-time recovery codes for the Owner and asks them to save them	Codes work once each; used codes are refused	P1
DEP-8	Owner lockout recovery: changing `SETUP_CODE` in the Cloudflare dashboard and redeploying re-opens a "reclaim owner" screen that needs the new code plus a confirmation sent to the Owner's email if email is set up	Reclaim works with the new code; old sessions are revoked; the audit log records it	P1
DEP-9	A Settings > Domain page explains how to add a custom domain in the Cloudflare dashboard and checks that the domain resolves to the app	Page reports pass or fail for a given hostname	P1
DEP-10	The Phase 0 deploy test runs on a brand-new Cloudflare account and a brand-new GitHub account, by someone other than the developer	Time from click to first saved quote is recorded and under 10 minutes	P0

6.2 Sign-in and sessions (AUTH)

ID	REQUIREMENT	DONE WHEN	PHASE
AUTH-1	Members sign in with a passkey (WebAuthn). No passwords exist anywhere in the product	A member with a passkey signs in on Chrome, Safari, Firefox, Edge, iOS, and Android	P0
AUTH-2	Members can register more than one passkey and remove old ones	Removing a passkey blocks sign-in with it on the next attempt	P1
AUTH-3	Email sign-in links work when an email provider is set up: one-time, 15-minute expiry, single use	A reused or expired link is refused	P1
AUTH-4	Optional sign-in with Google or Microsoft (OIDC), turned on by an Admin with their own client ID and secret	A Google Workspace user signs in and lands in the right workspace; users outside allowed domains are refused	P2
AUTH-5	Invites: an Admin creates an invite for an email and role. The app shows a copyable invite link and emails it when email is set up. Links expire after 7 days	Invitee registers a passkey from the link and gets the invited role	P1
AUTH-6	Sessions use an HttpOnly, Secure, SameSite=Lax cookie holding a random 256-bit token. The database stores only its SHA-256 hash	A database dump contains no usable session tokens	P0
AUTH-7	Sessions end after 12 hours idle or 30 days total, and on role change, member removal, or sign-out everywhere	Each case ends the session within one request	P1
AUTH-8	Members see their own active sessions and passkeys and can revoke any of them	Revoked session's next request returns 401	P1
AUTH-9	Sign-in, invite, and recovery endpoints are rate limited per IP and per account	20 failed attempts in 10 minutes locks that path for 15 minutes and logs it	P1
AUTH-10	Every sign-in path stays under 3 ms of Worker CPU	Measured in the free-plan load test (TST-6)	P0
AUTH-11	The admin guide shows how to put the whole app behind Cloudflare Access (free up to 50 users) for teams that want a second lock or an IP allowlist	Following the guide on a test account blocks visitors without an Access login, and the customer page path stays open	P2

6.3 Workspace, members, and branding (WS)

ID	REQUIREMENT	DONE WHEN	PHASE
WS-1	One deployment holds one workspace in the self-deployed edition. The hosted edition holds many (HST-1)	Self-deployed has no workspace switcher	P1
WS-2	Admins add, change the role of, and remove members. Removing a member keeps their quotes and reassigns ownership to a chosen member	Removed member can't sign in; their quotes show the new owner and the history keeps the original author	P1
WS-3	Roles and permissions follow Appendix A exactly	Generated permission tests pass (TST-3)	P1
WS-4	Branding: logo (SVG or PNG), primary and accent colors, one of 6 bundled fonts, company address block, footer text	Customer page, PDF, and emails all show the brand	P1
WS-5	Several selling entities (legal names, addresses, tax IDs, bank details) per workspace; each quote picks one	PDF and customer page show the chosen entity's details	P1
WS-6	Quote number format per workspace, for example `Q-{YYYY}-{SEQ:4}`. The Durable Object hands out numbers in one transaction, so two quotes never share a number	200 parallel creates produce 200 unique, gapless numbers	P1
WS-7	Work calendar: working days of the week and holidays, used by date math in formulas	`workdays_between` matches a hand-checked calendar for one year	P1
WS-8	Workspace settings change history: who changed what and when, with before and after values	Every settings save writes an audit entry	P1
WS-9	Entity attributes (for example travel or lodging multipliers per selling entity) usable in formulas as `entity.<attribute>`	A quote under an entity with a 1.2 travel multiplier prices travel 20% higher, and the trace shows the multiplier	P1

6.4 Customers, contacts, and sites (CUS)

ID	REQUIREMENT	DONE WHEN	PHASE
CUS-1	Customer companies with name, billing address, tax ID, tax-exempt flag, default currency, default language, payment terms, and notes	A quote for a customer picks up these defaults	P1
CUS-2	Contacts under a customer with name, title, email, phone; one marked primary	Quote recipient defaults to the primary contact	P1
CUS-3	Sites (service locations) under a customer with address and country. Country resolves to a pricing region (RATE-4)	Picking a site sets the quote's region and rates	P1
CUS-4	Installed assets per site: asset type, model, serial number, install date, custom attributes	A quote can pull a site's assets into an asset-based line group	P2
CUS-5	Customer price agreements: a customer-specific price list or discount percent per catalog category, with start and end dates	A quote for that customer uses the agreement price and the trace shows why	P2
CUS-6	Customer search by name, contact, email, or city; results in under 300 ms with 10,000 customers	Measured on the fixture data set	P1
CUS-7	Duplicate check on create by name and email domain	Creating a near-duplicate shows the existing record first	P2
CUS-8	Customer page lists every quote for that customer with status and value	List matches the quotes table filtered by customer	P1
CUS-9	Custom fields and tags on customers (for example industry or market), usable as report filters and in formulas as `customer.<field>`	A report filtered by a tag matches a direct count; a formula reads the field	P2

6.5 Catalog and assets (CAT)

ID	REQUIREMENT	DONE WHEN	PHASE
CAT-1	Catalog items: SKU, name, description, category, unit (each, hour, day, month, year, km, mi, or custom), price per currency, optional cost, billing frequency (one-time, monthly, quarterly, yearly), taxable flag, active flag	An item added to a quote shows the right unit, price, and frequency	P1
CAT-2	Custom attributes per category (number, text, choice, money), usable in formulas as `item.<attribute>`	A formula reading `item.pm_hours` returns the item's value	P1
CAT-3	Labor roles as a catalog type: hourly and daily rate per region, overtime and travel-time multipliers, and default lodging and per-diem flags	A labor line for a role prices from the role's regional rate	P1
CAT-4	Asset types (the general form of the prototype's machine list): attributes such as service hours per visit, replacement value, parts allowance percent	An asset-based template prices per asset from these attributes	P1
CAT-5	Bundles: a named set of items and quantities that expands into lines when added	Adding a bundle adds each line; removing it removes them together	P2
CAT-6	Price history: every price change keeps the old price with dates. Sent quotes keep the price they were sent with	Changing a price doesn't move any sent quote's total	P1
CAT-7	Catalog search by SKU, name, or attribute; 20,000 items return results in under 300 ms	Measured on fixture data	P1

ID	REQUIREMENT	DONE WHEN	PHASE
CAT-8	Items can be hidden from the customer as separate lines and rolled into a parent line	Customer page and PDF show the parent line only; the trace shows both	P2
CAT-9	Parts can be linked to asset types, so the builder offers an asset's parts list first	Picking an asset type shows its linked parts at the top of the parts search	P2

6.6 Rates, regions, currency, and tax (RATE)

ID	REQUIREMENT	DONE WHEN	PHASE
RATE-1	Rate tables: named rates (for example `technician_hourly`, `mileage`, `lodging_night`, `per_diem`) with a value per region, a unit, a currency, and an effective date	A formula `rate.lodging_night` returns the value for the quote's region and date	P1
RATE-2	Rate grid editor: rates as rows, regions as columns, inline edit, bulk paste from a spreadsheet, CSV import and export	Pasting a 50 x 10 block from Google Sheets saves 500 values in one action	P1
RATE-3	Regions with a default currency, distance unit (km or mi), and tax rule	A region in Poland quotes in PLN with km	P1
RATE-4	Country to region mapping seeded with all ISO countries; Admins assign countries to regions	Every country resolves to exactly one region or to "unassigned", which blocks quoting until set	P1
RATE-5	Exchange rates entered by hand per currency pair with a date. Optional daily fetch from a public FX source the Admin turns on; the source host shows on the outbound-hosts list	With fetch off, the app makes no FX calls	P1
RATE-6	Each sent revision freezes its exchange rates. The customer page can show a second currency using the frozen rate, with the rate and date printed	Changing today's rate doesn't move a sent quote	P1
RATE-7	Tax rules per region: rate, which categories are taxable, and display label (VAT, GST, sales tax). Tax-exempt customers skip tax	A taxable line in a 20% VAT region adds the right tax; an exempt customer pays none	P2
RATE-8	Rounding per currency (minor units) and an optional per-template rounding step for line totals (nearest 1, 5, 10, 50, 100)	Rounded totals match the configured step in every pricing example	P1
RATE-9	Distance for travel: typed in by the rep, or computed from the site address by a maps provider the Admin adds with their own key (Mapbox or Google)	With no key set, distance is a manual field and the app makes no maps calls	P2

6.7 Pricing engine (PRC)

ID	REQUIREMENT	DONE WHEN	PHASE
PRC-1	The engine is a pure TypeScript package with no I/O, used unchanged in the browser (live preview) and in the Worker (authoritative result)	Browser and Worker produce byte-identical results for every pricing example	P0
PRC-2	Money uses decimal arithmetic, never binary floats. Line totals round to the currency's minor unit, then to the template's rounding step	`0.1 + 0.2` style errors don't appear in any example; a property test of 100,000 random quotes finds no rounding drift between line sum and total	P0
PRC-3	A quote template (quote type) defines inputs, line groups, lines, options, add-ons, discount order, and which sections the customer sees	Each prototype quote type is rebuilt as a template with no code changes (G-2)	P0 to P1
PRC-4	Input types: number, integer, money, percent, yes/no, choice, multi-choice, date, date range, text, catalog pick (item and quantity), customer site, asset list, labor list (role, people, hours or days, lodging and per-diem flags), and repeating groups of any of these	The builder renders and validates each type	P1
PRC-5	Formula language: arithmetic, comparisons, and/or/not, `if`, `min`, `max`, `round(x, step)`, `ceil`, `floor`, `abs`, `clamp`, `sum(group, expr)`, `count(group, cond)`, `lookup(table, key)`, `coalesce`, `days_between`, `workdays_between`. No loops, no user functions, max 500 nodes per formula	The parser accepts the grammar in Appendix B and rejects everything else with a line and column	P0
PRC-6	Formulas compile to a stored syntax tree at publish time and run in an interpreter. The engine never uses `eval` or `new Function` (Cloudflare Workers block both)	A code search finds neither; CI fails if either appears	P0
PRC-7	Every value carries a unit (money in a currency, hours, days, distance, count, percent). The publish check rejects mixed-unit math such as hours plus money	A template adding hours to dollars fails to publish with a clear message	P1
PRC-8	Line definition: label, group, quantity formula, unit price formula, include condition, optional cost formula, billing frequency, taxable, discountable, and customer visibility (shown, rolled into parent, internal only)	Each field works in a pricing example	P0
PRC-9	Single-source quantities: a template can mark a quantity (for example room-nights) as owned by one line. Any other line that bills the same quantity has to reference it, and the publish check fails if two lines each bill it from their own inputs	A template that bills lodging on two lines from separate inputs can't publish	P1
PRC-10	Built-in blocks so most templates need no formulas: Labor, Travel (drive legs and fixed-fare legs), Lodging, Per diem, Asset-based service, Parts, Recurring service with term, Fixed fee, and Custom lines	A Pricing editor builds the field-service template from blocks alone	P1

ID	REQUIREMENT	DONE WHEN	PHASE
PRC-11	Options (good/better/best or named tiers): one quote computes every option side by side. Options change parameters such as discount percent, included hours, response-time fee percent	The customer page shows each option's total; the trace shows each option separately	P1
PRC-12	Optional add-ons the customer can tick on the quote page. Their amounts freeze at send time; ticking one adds a fixed amount and runs no formula in the customer's browser	Add-on toggles change the total by exactly the frozen amount	P1
PRC-13	Discounts at three levels (line, option, quote), each percent or fixed amount, applied in the order the template sets. The trace shows each step	Two templates with different orders produce the documented totals	P1
PRC-14	Rep overrides: a rep with permission can override a rate or price on one quote. The override shows in the trace with the original value and counts toward approval rules	An overridden quote lists every override on the approval screen	P1
PRC-15	Every computed number keeps a trace: formula, inputs with values, and result. The trace saves with each revision	Any total on a sent quote can be explained down to inputs from the saved trace alone	P0
PRC-16	Recurring pricing: monthly, quarterly, and yearly lines show per-period and term totals; the quote shows one-time and recurring subtotals apart	A 36-month contract shows monthly, yearly, and contract totals that agree	P1
PRC-17	Speed: the largest fixture quote (300 lines, 3 options) prices in under 2 ms of Worker CPU	Measured in workerd in CI	P0
PRC-18	The engine version is stored with each revision. A newer engine never changes the total of a stored revision without showing it	Opening an old revision under a new engine shows the stored total and flags any difference	P1

6.8 Pricing checks (CHK)

ID	REQUIREMENT	DONE WHEN	PHASE
CHK-1	Pricing examples: a saved set of inputs plus the expected total of every line and every option, attached to a template	An example runs and shows pass or fail per line	P1
CHK-2	Saving a template draft runs every example. Publishing is blocked while any example fails, unless a Pricing editor accepts each changed number with a reason	The audit log records each accepted change with the reason	P1
CHK-3	Change preview: before publishing, the editor sees which open (sent, not yet accepted) quotes would change on re-price and by how much	The preview lists quote number, old total, new total, and difference	P1
CHK-4	Built-in invariant checks on every example: no negative total unless the template allows it, no quantity billed twice (PRC-9), every customer-visible line has a trace, option totals equal the sum of their lines	A template that breaks an invariant fails with the line named	P1
CHK-5	"Save as example" from any real quote in the builder	One click creates an example from the current inputs and totals	P1
CHK-6	Starter templates ship with their examples, and CI runs them on every commit	CI fails on any change to a starter template's numbers	P0

6.9 Quote builder (QB)

ID	REQUIREMENT	DONE WHEN	PHASE
QB-1	Guided mode: one question per screen, with screens shown or hidden by earlier answers. Keyboard: Ctrl+Enter next, Shift+Enter back	A rep completes a field-service quote with the keyboard only	P1
QB-2	Full-form mode: every section on one scrolling page with a section index	Same quote completes in full-form mode with identical totals	P1
QB-3	Live total and a breakdown panel (Ctrl+B) showing the trace for the current inputs	The panel matches the server result after save	P1
QB-4	Autosave 2 seconds after the last change, at most once per 15 seconds, writing at most 2 database rows per save	Load test (TST-6) confirms the row count	P1
QB-5	If the connection drops, edits keep going into a local copy and sync on reconnect. Conflicts follow QB-6	Pulling the network for 5 minutes loses no edits	P2
QB-6	Optimistic locking: each save sends the version it started from. A stale save gets a conflict screen showing what changed and who changed it	Two editors on one quote never overwrite each other's work without a warning	P1
QB-7	Presence: the builder shows who else has the quote open	A second viewer appears within 2 seconds	P2
QB-8	Start a quote from scratch, from a template, by duplicating an earlier quote, or from a customer's last quote	Each path opens a draft with the right defaults	P1
QB-9	Cover letter and scope text with rich text (headings, bold, lists, links, tables) and merge fields such as `{customer.name}` and `{quote.valid_until}`	Merge fields fill on the customer page and PDF; unknown fields fail validation	P1
QB-10	Terms and conditions chosen from workspace terms blocks, editable per quote with permission	The sent revision freezes the exact terms text	P1
QB-11	Internal notes and internal attachments that customers never see	Neither appears in the customer page, PDF, emails, or customer API responses	P1

ID	REQUIREMENT	DONE WHEN	PHASE
QB-12	Customer attachments (spec sheets, drawings) up to 10 MB each, shown on the customer page	Customer downloads each attachment	P1
QB-13	Validity: valid-from and valid-until dates with a workspace default length	Customer page shows the dates; accept is blocked after expiry	P1
QB-14	The builder works on a phone-width screen	Full quote completes on a 390 px wide screen	P2
QB-15	Margin view for roles with cost access: cost, margin, and margin percent per line and per option	Sales without cost access never receive cost data from the API	P2
QB-16	Re-price: a rep re-runs a draft, or a copy of a sent revision, against today's rates and template version and sees old and new side by side before keeping it	The comparison lists every line whose amount changed; keeping it creates a new draft revision	P1

6.10 Quote lifecycle (QT)

ID	REQUIREMENT	DONE WHEN	PHASE
QT-1	Statuses: Draft, Pending approval, Approved, Rejected, Sent, Viewed, Accepted, Declined, Expired, Withdrawn. Allowed moves follow Appendix C	Any move outside Appendix C returns 409	P1
QT-2	One quote has one number and one customer link. Revisions sit under it. The customer link always shows the latest sent revision	A customer with an old link sees the newest revision and a "this quote was updated" note	P1
QT-3	Sending freezes the revision: inputs, lines, totals, trace, exchange rates, terms, template version, and engine version. A SHA-256 hash of the frozen revision is stored	Editing after send creates a new draft revision; the sent one never changes	P1
QT-4	Revising a sent quote creates a new draft revision. Sending it replaces the customer-facing revision and records the old one as superseded	Revision history shows every sent revision with its total	P1
QT-5	Revising after acceptance needs an Admin and creates a change-order revision that the customer accepts as its own step	The original acceptance record stays intact	P2
QT-6	Expiry: a Durable Object alarm moves quotes to Expired at the end of the valid-until day in the workspace time zone	A quote expires within 5 minutes of the deadline	P1
QT-7	Withdraw: the rep or an Admin can withdraw a sent quote. The link then shows "this quote is no longer available"	Accept is refused on a withdrawn quote	P1
QT-8	Status changes happen inside one Durable Object transaction with a version check, so two finalize or accept requests can't both win	100 parallel accept attempts produce exactly one acceptance	P1
QT-9	Quote list with filters (status, owner, customer, template, date, value) and saved views	Filters return the right results on 10,000 fixture quotes in under 500 ms	P1
QT-10	"Needs action" list per member: approvals waiting on them, accepted quotes not yet acknowledged, quotes expiring in 7 days, unanswered customer questions	Each item links to the quote and clears when handled	P1

6.11 Approvals (APR)

ID	REQUIREMENT	DONE WHEN	PHASE
APR-1	Approval rules per workspace or per template: total discount over X percent, margin under Y percent, total over Z, any rep override, or specific items present	A quote that meets any rule can't be sent until approved	P1
APR-2	Each rule names approvers by role or by person. The first approver to act decides	The approval screen shows who decided and when	P1
APR-3	Approvers see the reason the rule fired, the overrides, the discount steps, and the trace	All four appear on the approval screen	P1
APR-4	Approve or reject with a comment. Rejection returns the quote to Draft	The rep is notified with the comment	P1
APR-5	Any edit after approval clears the approval and re-runs the rules	Editing an approved quote returns it to Draft	P1
APR-6	Approvers can't approve their own quotes unless the workspace allows it	Self-approval is refused by default	P1
APR-7	Approval reminders after a set number of hours	Reminder reaches the approver in-app and by email if set up	P2

6.12 Customer quote page (CV)

ID	REQUIREMENT	DONE WHEN	PHASE
CV-1	The customer page opens at `/q/<token>` with a 128-bit random token, no sign-in. Responses carry `noindex` and `no-store`	Tokens can't be guessed; search engines don't index pages	P1

ID	REQUIREMENT	DONE WHEN	PHASE
CV-2	The page shows the brand, entity, cover letter, scope, options side by side, line groups with subtotals, add-ons, recurring and one-time totals, tax, validity, terms, attachments, and the rep's contact	Every section renders from the frozen revision	P1
CV-3	The page loads in under 1.5 seconds on a mid-range phone on 4G, with under 250 KB of JavaScript (compressed) before the PDF library loads	Lighthouse mobile run meets the target	P1
CV-4	Currency toggle between the quote currency and a second currency using the frozen rate, with the rate and date shown	Toggle changes every amount by the frozen rate	P1
CV-5	Accept: pick an option and add-ons, then enter name, title, email, PO number (optional), and a typed or drawn signature, and tick agreement to the terms	Acceptance saves all fields plus time, IP, user agent, and the revision hash	P1
CV-6	After acceptance, the page shows an acceptance certificate: who, when, which option, total, and the revision hash. The PDF includes the certificate page	Regenerating the PDF from the frozen revision gives the same hash on the certificate	P1
CV-7	Optional email check before accepting: a 6-digit code sent to the signer's email, when email is set up and the workspace turns it on	Accept is blocked until the code matches	P2
CV-8	Decline with an optional reason	The rep is notified with the reason; the quote moves to Declined	P1
CV-9	Questions: the customer posts a question on the page; the rep answers in the app; both see the thread	A question notifies the rep within 1 minute	P2
CV-10	View tracking: each open records time, country (from Cloudflare's request data, no outside lookup), and device type. Opens by signed-in members of the workspace don't count	Opening as a member adds no view; opening signed out adds one	P1
CV-11	Expired quotes show "request an updated quote", which notifies the rep	The request appears in the rep's Needs action list	P1
CV-12	Optional access code per quote for customers who want an extra lock	Without the code the page shows only the company name and a code field	P2
CV-13	Print stylesheet so browser print gives a clean document	Printing from Chrome and Safari matches the PDF layout section for section	P1
CV-14	Link preview image (for email and chat apps) comes from the workspace brand and is generated in the Admin's browser when branding is saved. No server-side rendering	Pasting a quote link in Slack shows the brand card	P2
CV-15	Content blocks: a template can include sections with text, images, and cards and no price, such as a product overview or hosting options	Content blocks render on the customer page and PDF and never change a total	P1

6.13 PDF documents (PDF)

ID	REQUIREMENT	DONE WHEN	PHASE
PDF-1	PDFs render in the browser from the same render model as the customer page. The PDF library loads only when a PDF is requested	No server-side browser is used anywhere	P1
PDF-2	Layout: cover with brand and entity, letter, scope, options, line groups, totals, terms, attachments list, acceptance certificate when accepted	A 300-line quote renders on Letter and A4 with no clipped or overlapping content	P1
PDF-3	The PDF footer prints quote number, revision, page x of y, and the revision hash	Every page carries the footer	P1
PDF-4	The same frozen revision always produces the same PDF content	Two renders of one revision match text-for-text	P1
PDF-5	Per-workspace PDF layout choices: page size, logo position, show or hide unit prices, show or hide line detail	Each choice changes the output as described	P2
PDF-6	Internal PDF with cost and margin for roles with cost access	Internal PDF is never attached to customer emails	P2

6.14 Email (EML)

ID	REQUIREMENT	DONE WHEN	PHASE
EML-1	With no provider set up, "Send" opens the rep's own email app with the customer, subject, and a message holding the quote link, and marks the quote Sent	The quote moves to Sent; the link works	P1
EML-2	Provider adapters: Cloudflare Email Service (on a Workers Paid plan), Resend, and SMTP2GO. Postmark and Amazon SES follow	A test send from Settings > Email succeeds on each provider	P1 to P2
EML-3	Email templates for quote sent, reminder, approval needed, accepted, declined, question, invite, and sign-in link, with merge fields and per-language versions	Each template previews with sample data before saving	P1
EML-4	Sending domain setup page shows the DNS records the provider needs and checks them	Page reports pass or fail per record	P2
EML-5	Follow-up reminders: the rep schedules reminders (for example 3 and 7 days after send if not viewed or not accepted). A Durable Object alarm sends them	Reminder stops once the customer accepts or declines	P2

ID	REQUIREMENT	DONE WHEN	PHASE
EML-6	Every outgoing email goes through the outbox (INT-6) with retries and a delivery log	A provider outage delays sends; nothing is lost	P1
EML-7	Test mode: all email goes to one address the Admin sets	No customer receives mail while test mode is on	P1

6.15 Notifications and activity (NTF)

ID	REQUIREMENT	DONE WHEN	PHASE
NTF-1	In-app notifications pushed over one WebSocket per open tab, using the Durable Object hibernation API. No polling	A notification shows in under 2 seconds; an idle tab costs no Durable Object duration	P1
NTF-2	Events: approval needed, approved, rejected, quote viewed (first view and after a gap of 24 hours), accepted, declined, question asked, expiring soon, expired, mentioned in a note	Each event reaches the right people	P1
NTF-3	Per-member preferences: in-app, email, both, or off, per event	Preferences take effect on the next event	P2
NTF-4	Activity timeline on each quote: every status change, view, email, approval, comment, and revision with time and actor	Timeline matches the audit log for that quote	P1
NTF-5	Slack and Microsoft Teams channel alerts by incoming webhook URL, per event	A test alert posts to the channel	P4

6.16 Reports (RPT)

ID	REQUIREMENT	DONE WHEN	PHASE
RPT-1	Dashboard: quotes and value by status, sent this month, accepted this month, win rate by count and by value, average days from send to accept	Numbers match a hand count over the fixture data	P2
RPT-2	Breakdowns by rep, template, customer, region, and option picked	Each breakdown totals to the dashboard figure	P2
RPT-3	Expiring soon and stale (sent, not viewed in 7 days) lists	Lists match the quote data	P2
RPT-4	Reports read from summary rows updated on each status change, never from full scans	A report load reads fewer than 5,000 rows on the 10,000-quote fixture	P2
RPT-5	CSV export of any report	Export opens in Excel and Google Sheets with the right columns	P2
RPT-6	Test data exclusion: quotes flagged as test never count in reports	Flagging a quote as test removes it from every figure	P2

6.17 Integrations and API (INT)

ID	REQUIREMENT	DONE WHEN	PHASE
INT-1	REST API at `/api/v1` with scoped API keys (read quotes, write quotes, read catalog, write catalog, read reports). Keys are shown once and stored hashed	A key without write scope gets 403 on writes	P4
INT-2	OpenAPI document for `/api/v1`, published with each release	Generated client compiles and passes the API tests	P4
INT-3	Webhooks for quote created, sent, viewed, accepted, declined, expired, and approval events, signed with HMAC-SHA256 and a timestamp	A receiver verifies the signature with the documented recipe	P4
INT-4	HubSpot: link a quote to a deal, push line items and the quote link, move the deal stage on accept	Accepting a test quote moves the test deal's stage	P4
INT-5	Salesforce: link a quote to an Opportunity, fill customer fields from the Account, post status events as Activities, set the stage on accept	Same as INT-4 on a Salesforce developer org	P4
INT-6	Outbox: webhooks, CRM calls, and emails go into an outbox table and a Durable Object alarm delivers them with retries at 1 minute, 5 minutes, 30 minutes, 2 hours, and 12 hours, then marks them failed	Admin > Deliveries shows each attempt; a failed one can be retried by hand	P1
INT-7	Each outbox run makes at most 40 outbound requests, inside the free plan's 50 per invocation	A backlog of 500 items drains over several alarm runs	P1
INT-8	Integration secrets are encrypted at rest with AES-GCM using a key from a Worker secret	A database dump holds no readable integration tokens	P4

6.18 Instant-estimate widget (WID)

ID	REQUIREMENT	DONE WHEN	PHASE
WID-1	A script tag the customer adds to their own website renders an estimate form in a shadow DOM, styled by data attributes and CSS custom properties	Host page CSS doesn't change the widget; the widget doesn't change the host page	P4
WID-2	The Pricing editor marks which template inputs are public and how the estimate shows: exact, rounded, or a range	The widget asks only public inputs	P4
WID-3	The estimate is computed on the server. Formulas and rates never reach the visitor's browser	The widget's network responses contain only the display amount	P4
WID-4	Turnstile is required on submit. The widget stays off until an Admin pastes a Turnstile site key and secret	Without keys the widget shows nothing and the endpoint returns 403	P4
WID-5	Submitting creates a customer, a contact, and a draft quote with the visitor's answers, and notifies the assigned rep	The draft opens in the builder with the answers filled	P4
WID-6	Daily cap on widget estimates (default 500) and per-IP rate limit, so the widget can't use up the free plan	Calls past the cap return a friendly "try again tomorrow" message	P4

6.19 Import, export, and backup (IMP, BAK)

ID	REQUIREMENT	DONE WHEN	PHASE
IMP-1	CSV import for catalog, rates, customers, contacts, sites, and assets. The browser parses the file, shows a preview with per-row errors, and sends rows in batches of 500	A 20,000-row catalog imports without hitting the daily write limit warning	P1
IMP-2	Column mapping screen with saved mappings	Re-importing the same layout needs no remapping	P2
IMP-3	Full workspace export: one ZIP with JSON for every table plus files, downloadable by the Owner	The ZIP imports into an empty deployment and reproduces every quote, total, and hash	P1
IMP-4	The same export format moves a workspace between the self-deployed and hosted editions in either direction	Round trip hosted to self-deployed to hosted keeps every record	P3
BAK-1	Point-in-time restore from the Durable Object's built-in history, to any minute in the last 30 days, Owner only	Restoring to before a bulk delete brings the data back	P1
BAK-2	Before each schema migration, the app records a restore point and shows it in Settings > Backups	The restore point appears with the version it belongs to	P1
BAK-3	Optional nightly export to R2 for longer retention, off by default because R2 needs a payment method on the Cloudflare account	With R2 bound and turned on, a nightly ZIP appears in the bucket	P2

6.20 Updates (UPD)

ID	REQUIREMENT	DONE WHEN	PHASE
UPD-1	The repo ships `.github/workflows/update.yml`. In the customer's copy it runs weekly and on demand, fetches the latest release tag from the upstream repo, and opens a pull request	A test copy receives an update pull request within one run	P1
UPD-2	An Owner setting switches the workflow to merge the update on its own when CI passes	With the setting on, a release reaches the test copy with no clicks	P1
UPD-3	Merging deploys through Workers Builds. If Phase 0 finds that pushes from the workflow don't start a build, the workflow deploys with Wrangler and a Cloudflare API token the Owner adds as a GitHub secret	Either path deploys the update	P0 to P1
UPD-4	Schema migrations live in code and run inside the Durable Object on the first request after deploy, inside one transaction, forward only	Upgrading across 5 releases on fixture data succeeds	P1
UPD-5	Downgrade guard: if the database schema is newer than the code, the app shows "restore or update" and makes no writes	Deploying an old release over a new database writes nothing	P1
UPD-6	Settings > About shows the running version and release notes. The check for newer releases is off by default; when on, it calls the GitHub releases API, listed on the outbound-hosts page	With the check off, the app makes no calls to GitHub	P1
UPD-7	Releases follow semantic versioning with a changelog. Breaking changes ship only in major versions with a migration note	Each release has notes listing migrations	P1

6.21 Hosted edition (HST)

ID	REQUIREMENT	DONE WHEN	PHASE
HST-1	Each hosted workspace is its own Durable Object with its own SQLite database. A D1 database holds only the directory: workspaces, plans, domains, and billing state	No query can read across workspaces; the directory holds no quote data	P3
HST-2	Sign-up creates a workspace and the Owner's passkey in one flow, then runs the setup wizard (DEP-6)	New sign-up reaches an empty dashboard in under 3 minutes	P3
HST-3	Plans and billing through Stripe Checkout and the Stripe customer portal. Stripe webhooks set the plan; failed payments get a 14-day grace period, then read-only	Each state change in Stripe shows in the workspace within 1 minute	P3
HST-4	Plan limits (members, quotes per month, storage) are enforced with clear messages and never delete data	Going over a limit blocks new items only	P3
HST-5	Custom domains through Cloudflare for SaaS custom hostnames: the customer adds a CNAME and a TXT record; the app checks them and TLS is issued	A test domain serves the workspace over HTTPS	P3
HST-6	Managed email from the workspace's own domain after DNS verification, or from a shared amplifi sending domain by default	Both paths deliver to a test inbox with SPF and DKIM passing	P3
HST-7	Data location: workspaces can pick EU data location at sign-up, which creates their Durable Object in the EU jurisdiction	An EU workspace's object reports the EU jurisdiction	P3
HST-8	Support access: amplifi staff can enter a workspace only after the Owner grants time-limited access (1 to 72 hours). Every action during access is logged and visible to the Owner	Without a grant, staff tools return 403	P3
HST-9	Workspace export (IMP-3) and full deletion are self-serve for the Owner. Deletion removes the Durable Object's data and the directory record within 30 days	A deleted test workspace's object reports zero rows	P3
HST-10	amplifi staff console: workspace list, plan, usage, status, and support-access grants. No quote content is visible without a grant	Staff console screens contain no customer quote data	P3
HST-11	Legal pages for the hosted edition: terms of service, privacy policy, data processing terms, and a list of subprocessors (Cloudflare, Stripe, the email provider)	Pages are reachable from the sign-up form and every footer	P3

6.22 Usage and administration (ADM)

ID	REQUIREMENT	DONE WHEN	PHASE
ADM-1	Settings > Usage shows today's requests, rows written, and storage against the Cloudflare free-plan limits, counted by the app itself	Figures are within 10% of Cloudflare's own dashboard over one day	P1
ADM-2	Warnings to Admins at 70% and 90% of any daily limit, once per day each	A load test past 70% raises one warning	P1
ADM-3	Outbound-hosts page lists every host the app can call, with the setting that turns each one on. All are off by default	With defaults, a network capture over a full workday shows zero outbound calls (TST-8)	P1
ADM-4	Audit log viewer with filters by member, action, and date, and CSV export. The log is append-only and kept for 1 year	No API path updates or deletes an audit row	P1

6.23 Languages and accessibility (I18N, A11Y)

ID	REQUIREMENT	DONE WHEN	PHASE
I18N-1	Every customer-facing string (page, PDF, emails) comes from a label set with per-workspace overrides. Lookup order: workspace override, built-in translation, English	Overriding a label changes it everywhere it shows	P1
I18N-2	Built-in customer-facing languages: English at P1; Italian, Polish, French, German, Dutch, Portuguese, and Spanish at P2	A quote in each language shows no English strings	P1 to P2
I18N-3	Each quote has its own language; numbers, dates, and currency format by locale through the browser's `Intl` APIs	A German quote shows `1.234,56 €` style formatting	P1
I18N-4	The app interface ships in English. Interface translation files are in the repo for contributors	English covers every interface string	P1
A11Y-1	Builder, customer page, and admin screens meet WCAG 2.2 AA	axe finds no serious or critical issues; a keyboard-only pass completes every main flow	P1 to P2
A11Y-2	Reduced-motion setting is respected; no animation carries meaning	With reduced motion on, no transitions run	P1

6.24 Optional extras (EXT)

ID	REQUIREMENT	DONE WHEN	PHASE
EXT-1	Pay on accept: an Admin connects their own Stripe account; the customer pays a deposit (percent or fixed) or the full amount when accepting	A test-mode payment marks the acceptance paid	P5
EXT-2	Draft from a request: a rep pastes an email or RFQ text; the app fills the template inputs it can find and marks each as suggested for the rep to confirm	Suggested fields are highlighted until confirmed; nothing sends without the rep	P5
EXT-3	EXT-2 uses Workers AI within its free daily allowance, or the workspace's own AI provider key. It's off by default, shows on the outbound-hosts page, and has a monthly cap	With it off, no AI calls happen	P5
EXT-4	Starter template library: field service and repair, preventive maintenance plan with tiers, extended warranty, training and consulting, equipment with installation, parts order, IT managed services, and agency retainer	Each installs with its pricing examples passing	P1 to P2

07 Architecture

7.1 Stack

LAYER	CHOICE	REASON
Language	TypeScript everywhere	One language for engine, server, and browser
Server	Hono on Cloudflare Workers	Small, typed routing that runs unchanged in workerd for tests
Data	SQLite inside one Durable Object per workspace, through Drizzle's durable-sqlite driver	Real transactions (`transactionSync`), 30-day point-in-time restore on the free plan, 10 GB per workspace, and one writer per workspace, which removes the race conditions the prototype had to fix with advisory locks
Directory (hosted only)	D1	Cross-workspace lookups: domains, plans, billing
Live updates	WebSocket hibernation API on the workspace Durable Object	Push instead of polling; no duration charge while idle
Scheduled work	Durable Object alarms	Expiry, reminders, outbox delivery, daily usage counts. Uses none of the 5 free Cron Triggers
Front end	React with Vite, served as Worker static assets in single-page-application mode	Static asset requests are free and unlimited
PDF	pdfmake in the browser, loaded on demand	No server-side browser
Sign-in	SimpleWebAuthn (works on Workers), email links, optional OIDC	No password hashing
Validation	Zod 4.5 or later	Shared schemas for API and forms; Cloudflare's memory guidance names versions before 4.5
Money	A decimal library (decimal.js-light or big.js), wrapped in a `Money` type	No float errors in totals
Tests	Vitest with `@cloudflare/vitest-pool-workers`, Playwright for end-to-end	Tests run inside workerd, the real runtime
License	Source-available, terms in decision D-1	Lets anyone self-deploy, blocks resale of a hosted copy

7.2 Request path

```
Browser
  |-- static assets (free, cached at the edge)
  |-- /api/*  ->  Worker: session check, rate limit, route
  |                 |-> Workspace Durable Object (SQLite)
  |                 |     pricing engine, quote state, audit log,
  |                 |     outbox, alarms, WebSocket hub
  |                 |-> email / CRM / webhook calls only from the outbox
  |-- /q/<token>  ->  Worker -> Workspace DO (read frozen revision)
  |-- /ws         ->  Worker -> Workspace DO (hibernating WebSocket)
```

The Worker stays thin: it checks the session cookie (a SHA-256 lookup), applies the rate limit, and forwards to the workspace object. All reads and writes for a workspace happen inside its Durable Object, one request at a time.

7.3 Repository layout

```
/                       wrangler.jsonc, package.json, .dev.vars.example
/packages/engine        pricing engine: parser, unit checker, interpreter, money, trace
/packages/schema        Drizzle tables, migrations, Zod schemas
/packages/templates     starter templates + their pricing examples
/apps/worker            Hono routes, Durable Object class, adapters
/apps/web               React app: builder, admin, customer page, PDF
/tests                  API, permission, engine, parity, load, and e2e suites
/.github/workflows      ci.yml, release.yml, update.yml
/docs                   spec, admin guide, template authoring guide
```

The Deploy to Cloudflare button needs the app at the repo root with its own dependencies, so this is one package workspace with a single root `package.json` build.

7.4 Adapters

The app talks to the outside through interfaces with Cloudflare versions first: `Store` (workspace SQLite), `Directory` (D1, hosted only), `Mail` (none, Cloudflare Email Service, Resend, SMTP2GO), `Files` (SQLite blobs, optional R2), `Clock`, `Live` (WebSocket hub), and `Secrets`. Tests use in-memory versions of each.

7.5 Data model

All tables live in the workspace database unless marked. Money columns store integers in minor units plus a currency code. Every table has `id`, `created_at`, `updated_at`, and `created_by` where it applies.

TABLE	HOLDS
`settings`	Workspace name, brand, defaults, number format, calendar, feature switches
`members`	Name, email, role, status, last sign-in
`passkeys`	Credential id, public key, counter, device label, member
`sessions`	Token hash, member, created, last seen, expires, user agent
`invites`	Email, role, token hash, expires, used
`entities`	Selling entities: legal name, address, tax id, bank details
`customers`	Company record, defaults, tax-exempt flag

TABLE	HOLDS
`contacts`	People under a customer
`sites`	Service locations with country and resolved region
`assets`	Installed assets per site with attribute values (JSON)
`catalog_items`	Items, labor roles, asset types; attributes as JSON validated by category schema
`item_prices`	Price per currency with effective dates
`regions`	Currency, distance unit, tax rule
`country_regions`	ISO country to region
`rates`, `rate_values`	Named rates and their per-region dated values
`fx_rates`	Currency pair, rate, date, source
`tax_rules`	Region, rate, taxable categories, label
`templates`, `template_versions`	Quote types; each published version is immutable and holds the compiled syntax tree
`pricing_examples`	Inputs and expected outputs per template version
`quotes`	Number, customer, owner, status, current revision, customer token, validity
`quote_revisions`	Inputs, computed lines, totals, trace, frozen rates and terms, template and engine version, hash, sent time
`quote_events`	Status changes and activity for the timeline
`acceptances`	Signer, option, add-ons, PO, signature, time, IP, user agent, revision hash
`views`	Quote, time, country, device type
`comments`	Internal notes and customer questions, with visibility
`files`	Attachments (SQLite blobs up to 1.9 MB each in chunks, or R2 keys when R2 is bound)
`approvals`	Rule fired, approver, decision, comment
`notifications`	Member, event, read state
`outbox`	Kind, payload, attempts, next attempt, last error
`webhooks`, `api_keys`	Endpoints and hashed keys with scopes
`audit_log`	Append-only: actor, action, target, before and after
`report_daily`	Summary counts and values per day, rep, template, status
`usage_daily`	Requests, rows read and written, storage, per day
`drafts`	Per-member unsaved builder state
(Directory, D1) `workspaces`, `domains`, `subscriptions`	Hosted edition only

Indexes go on every foreign key and every list filter. Each write path is reviewed for rows written, because each index adds a billed row.

7.6 Pricing engine

The engine takes a template version, inputs, and a rate context, and returns lines, option totals, add-on amounts, tax, and a trace.

1. **Parse** formulas into syntax trees at publish time. The grammar is in Appendix B.
2. **Check units.** Each node gets a unit; mixing units fails the publish.
3. **Order.** Lines and named values form a dependency graph; a cycle fails the publish.
4. **Evaluate** with an interpreter over the stored trees, using decimal arithmetic.
5. **Round** each line to the currency's minor unit and the template's step.
6. **Discount** in the template's order, recording each step.
7. **Trace**: every node records its inputs and output, so any number can be explained.

The engine has no I/O and no clock. The caller passes in rates, exchange rates, the calendar, and "today". The browser and the Worker run the same package and have to agree to the byte (PRC-1).

7.7 Quote templates

A template is a versioned JSON document. A trimmed field-service example:

```
{
  "key": "field_service",
  "name": "Repair and field service",
  "inputs": [
    { "id": "techs", "type": "integer", "label": "Technicians", "min": 1, "default": 1 },
    { "id": "onsite_days", "type": "integer", "label": "Days on site", "min": 1 },
    { "id": "drive_km", "type": "number", "label": "Round-trip distance", "unit": "distance" },
    { "id": "lodging", "type": "yesno", "label": "Overnight stay", "default": true }
  ],
  "values": {
    "room_nights": "if(lodging, techs * max(onsite_days - 1, 0), 0)"
  },
  "groups": [
    { "id": "labor", "label": "Labor", "lines": [
      { "label": "Technician time", "qty": "techs * onsite_days * 8", "unit": "hour",
        "price": "rate.technician_hourly" } ] },
    { "id": "travel", "label": "Travel", "lines": [
      { "label": "Mileage", "qty": "drive_km", "unit": "distance", "price": "rate.mileage" },
      { "label": "Lodging", "qty": "room_nights", "unit": "night", "price": "rate.lodging_night",
        "owns": "room_nights", "when": "lodging" },
      { "label": "Per diem", "qty": "techs * onsite_days", "unit": "day", "price": "rate.per_diem" } ] }
  ],
  "discountOrder": ["line", "option", "quote"]
}
```

The `owns` field enforces PRC-9: no other line can bill `room_nights` on its own. That rule targets a bug class the prototype hit: lodging nights billed from two separate inputs.

7.8 Carrying over the prototype's arithmetic

The prototype's math lives in a 12,500-line builder file plus a 1,989-line breakdown module. The plan:

1. Export a fixture set from the prototype: inputs and stored totals for every finalized quote type, with customer names, addresses, and contacts replaced by fakes. The fixture keeps only numbers and structure.
2. Rebuild each prototype quote type as a template.

3. Run every fixture through the new engine. Totals have to match the prototype to the cent, except where the prototype has a known, documented defect; those differences are listed with the reason and the corrected value.
4. Keep the fixtures as private regression tests. They never enter the public repo.

7.9 Free-plan budget

Cloudflare's free plan limits (read October 20, 2026): 100,000 Worker requests a day, 10 ms CPU per request, 50 outbound requests per invocation, and for SQLite-backed Durable Objects 100,000 requests a day, 13,000 GB-seconds of duration a day, 5 million rows read and 100,000 rows written a day, and 5 GB of storage.

Estimate for one heavy user-day (3 hours in the builder, autosave at most every 15 seconds while editing, about half that time editing): 610 requests and 780 rows written.

PEOPLE QUOTING ALL DAY	REQUESTS (OF 100,000)	ROWS WRITTEN (OF 100,000)
25	15%	20%
50	30%	39%
100	61%	78%
128	78%	100%

One workspace object active for a full 24 hours uses 10,800 GB-seconds, under the 13,000 free. A self-deployed workspace can't run out of duration on its own.

These are estimates. Phase 2 replaces them with a measured load test (TST-6), and the usage page (ADM-1) shows live counts.

Design rules that keep this true:

- No polling. One hibernating WebSocket per tab.
- No per-request logging to the database. The audit log records actions, not page loads. Request logs go to Workers Logs (200,000 a day free, kept 3 days).
- Autosave writes at most 2 rows.
- Imports batch in the browser.
- Reports read summary rows.
- Outbox runs make at most 40 outbound calls each.

7.10 Deploy button contract

ITEM	SETTING
Button URL	`https://deploy.workers.cloudflare.com/?url=<public repo URL>`
Resources	Worker + static assets + one SQLite Durable Object class, declared in `wrangler.jsonc`
Secrets	`SETUP_CODE` (required); `RESEND_API_KEY`, `SMTP2GO_API_KEY`, `TURNSTILE_SECRET`, `INTEGRATION_KEY` (optional), listed in `.dev.vars.example` with descriptions in `package.json`
Build	`npm run build` (Vite + Worker bundle)
Deploy	`npm run deploy` = `wrangler deploy`; Durable Object migrations declared in `wrangler.jsonc`
Requirements on the deployer	A Cloudflare account and a GitHub or GitLab account. No payment method

ITEM	SETTING
Known limits	Repo has to be public, on github.com or gitlab.com, Workers only (not Pages). Builds: 3,000 minutes a month free, 1 at a time, 20-minute timeout

7.11 Hosted edition layout

One Worker serves every hosted workspace. The hostname (custom domain or `<slug>.quotes.amplifi.studio`) maps to a workspace id through the D1 directory, cached in memory per isolate. Each workspace has its own Durable Object, named by workspace id. Nothing in the directory holds quote data. Stripe webhooks update the directory. Custom domains use Cloudflare for SaaS (100 hostnames included, then $0.10 each a month).

08 Security

ID	REQUIREMENT	DONE WHEN	PHASE
SEC-1	Every API route checks the member's role inside the workspace Durable Object, not only in the Worker	Calling the Durable Object straight from a test harness with a valid session but the wrong role returns 403 (TST-3)	P1
SEC-2	Every state-changing request checks the `Origin` header against the deployment's hosts, on top of SameSite cookies	A cross-site form post is refused	P1
SEC-3	Customer-page responses go through an allowlist serializer: only fields marked customer-safe leave the server	A test that adds a new internal field to the quote finds it absent from every customer response	P1
SEC-4	Content Security Policy with no inline scripts, no `eval`, and `frame-ancestors` set to none except on the widget script	Mozilla Observatory grade A or better on the customer page	P1
SEC-5	Rich text (letters, scope, terms, comments) is sanitized on the server to an allowlist of tags and attributes	Known XSS payloads stored in each field render as text	P1
SEC-6	Uploads: size limit, type checked from file contents, served with `Content-Disposition: attachment` except images. SVG logos are sanitized before storage	An SVG with a script tag stores without the script	P1
SEC-7	Formula limits: 500 nodes per formula, 50,000 evaluation steps per quote, 1 MB of inputs per quote. Over the limit, the save fails with a message	A crafted template can't push a save past 5 ms of CPU	P0
SEC-8	Token and code comparisons use constant-time compare	Code review checklist and a lint rule cover it	P1
SEC-9	Rate limits that matter for safety (sign-in, setup code, accept, widget) count inside the Durable Object, because Cloudflare's rate-limit binding counts per location and can lag	50 requests from different locations still hit the limit	P1
SEC-10	Webhook and integration URLs have to be `https` with a public hostname; IP literals are refused	An `http://` or IP-literal URL fails validation	P4
SEC-11	Secrets never appear in logs, errors, exports, or API responses. Settings screens show only the last 4 characters	A search of a day of Workers Logs finds no secret values	P1
SEC-12	Dependencies are pinned with a lockfile; CI runs `npm audit` and fails on high or critical findings; a license check blocks GPL and AGPL code	CI blocks a test PR that adds a GPL package	P0
SEC-13	Security events (sign-in, failed sign-in, passkey added or removed, role change, export, restore, API key created, support access) write to the audit log	Each event appears in the audit log with actor and IP	P1
SEC-14	Before the hosted edition opens, an outside reviewer tests it. The prototype's security-test findings (open CORS, missing ownership checks, non-constant-time compares, injection through string-built queries, missing authorization on status changes) form the starting checklist	Every finding is fixed or accepted in writing before launch	P3
SEC-15	`SECURITY.md` in the repo with a reporting address and a 90-day disclosure window	File present and linked from the README	P1

09 Privacy and legal

- **Self-deployed edition.** The customer runs the app in their own Cloudflare account and controls all data. amplifi receives nothing: no telemetry, no error reports, no update pings unless the Owner turns on the version check (UPD-6). Cloudflare is the customer's own provider under the customer's own Cloudflare terms.

- **Hosted edition.** ACWebDev, LLC processes workspace data for the workspace Owner. The hosted terms include a data processing agreement and a subprocessor list (Cloudflare, Stripe, the email provider). Owners can export and delete their data at any time (HST-9).
- **Personal data kept.** Members: name, email, passkeys, sessions. Customer contacts: name, title, email, phone. Signers: name, title, email, IP address, user agent, signature image or typed name. Views store country and device type only, no IP address.
- **Retention.** Quotes and acceptances stay until the workspace deletes them. Audit log: 1 year. Sessions: deleted 30 days after expiry. Workers request logs: 3 days (Cloudflare's free-plan retention).
- **Electronic acceptance.** The accept flow records the signer's identity, intent (typed name or drawing plus a terms checkbox), time, IP address, and the hash of the exact revision shown. US law (ESIGN Act and UETA) and the EU's eIDAS rules for simple electronic signatures usually accept records like this, but rules differ by country and contract type. A lawyer reviews the accept flow and hosted terms before the hosted edition opens. This document isn't legal advice.
- **Cookies.** The app sets one session cookie for members and none for customers viewing a quote. No consent banner is needed for the app itself; the hosted marketing site handles its own.

10 Editions and plans

	SELF-DEPLOYED (FREE)	HOSTED TEAM	HOSTED BUSINESS
Price	$0	$49 per month	$149 per month
Where it runs	Customer's own Cloudflare account	amplifi's Cloudflare account	amplifi's Cloudflare account
Members	Unlimited	Up to 10	Unlimited
Features	All	All except SSO and EU data location	All
Custom domain	Customer adds it in Cloudflare	Included	Included
Email sending	Customer's provider key, or the rep's own email app	Included (shared domain or their own)	Included
Backups	30-day restore (built in)	30-day restore plus nightly export	Same, plus 1-year export retention
Support	GitHub issues	Email, 2 business days	Email, 1 business day
Setup service	Quoted per job	Quoted per job	Quoted per job

Setup service: amplifi builds the customer's templates, catalog, rates, and pricing examples from their price sheets.

11 Costs

Self-deployed. $0 for typical teams. Costs appear only if the customer chooses them: Cloudflare Workers Paid ($5 a month) above the free limits or to send email through Cloudflare; an email provider past its free tier (Resend free: 3,000 a month, 100 a day; SMTP2GO free: 1,000 a month, 200 a day); R2 for long-term exports, which needs a payment method on the Cloudflare account even inside the free allowance.

amplifi, for the free edition. $0. The public repo, the button, and the update workflow cost nothing to run.

amplifi, for the hosted edition. Workers Paid at $5 a month includes 10 million requests, 30 million CPU milliseconds, 1 million Durable Object requests, 400,000 GB-seconds of Durable Object duration, 50 million rows written, and 3,000 emails a month. Estimate for 100 workspaces with 5 heavy users each on 22 workdays:

ITEM	PER MONTH
Workers Paid base plan	$5.00
Durable Object duration*	$106.38
Durable Object requests	$0.86
CPU	$0.21

ITEM	PER MONTH
Email (15,000 sent)	$4.20
Total	**$116.65**
Per workspace	$1.17

* Worst case: assumes each workspace's object stays awake 9 hours every workday. Hibernating WebSockets should cut this; the Phase 2 load test measures it.

Custom domains: 100 included, then $0.10 each a month. Stripe fees apply per payment at Stripe's standard rates.

12 Testing

ID	REQUIREMENT	DONE WHEN	PHASE
TST-1	Engine unit tests for every function in Appendix B plus property tests (random quotes, line sum equals total, rounding stable)	100% branch coverage on the engine package	P0
TST-2	Prototype parity suite: anonymized prototype quotes rebuilt under the new templates match the prototype's stored totals to the cent, or appear on a documented-differences list with the reason	Every fixture passes or is listed	P0 to P1
TST-3	Permission tests generated from Appendix A: every action by every role, expected allow or deny	A change to Appendix A without matching code fails CI	P1
TST-4	API tests run inside workerd through `@cloudflare/vitest-pool-workers`	Suite passes on every pull request	P0
TST-5	End-to-end tests in Chrome, Safari (WebKit), and Firefox: build a quote, send it, open the customer page, accept, download the PDF	Suite passes on every release	P1
TST-6	Free-plan load test: replay 50 heavy user-days against a free account, then measure requests, rows read and written, Durable Object duration, and Worker CPU	Every endpoint's 99th-percentile CPU stays under 8 ms; daily totals stay under 50% of each free limit	P2
TST-7	Deploy test on a brand-new Cloudflare account and GitHub account by someone other than the developer, every minor release	Click to first saved quote in under 10 minutes	P0, then every release
TST-8	Zero-outbound test: a network capture over a scripted workday with default settings	Zero outbound connections	P1
TST-9	Upgrade test: install the last 5 releases in order on fixture data, then the new one	All migrations run; totals and hashes unchanged	P1
TST-10	Browser and Worker engine parity on every pricing example	Byte-identical results	P0
TST-11	Accessibility checks with axe on every screen in CI, plus a manual keyboard and screen-reader pass per release	No serious or critical axe findings	P1
TST-12	Customer page performance budget in CI (CV-3)	Lighthouse mobile score 90 or above	P1
TST-13	Two blind reviews of the customer page and PDF by people outside the build, before each major release	Both reviews' findings fixed or accepted in writing	P1

13 Phases

PHASE	WEEKS	SCOPE	EXIT TEST
P0	1	Repo, `wrangler.jsonc`, Durable Object with SQLite, setup code and passkey Owner, engine core (parser, interpreter, money, trace), field-service template, 20 or more prototype fixtures, Deploy button	Fresh-account deploy under 10 minutes by a non-developer; every fixture matches the prototype to the cent or is a documented defect. Either test can stop the project
P1	6 to 9	Settings-based pricing (catalog, rates, regions, templates, blocks, options, discounts, pricing checks), builder, lifecycle, approvals, customer page with accept, PDF, email adapters, notifications, export and restore, update workflow, usage page, English labels	Self-deployed v1.0 released as a public repo; TST-1 to TST-5 and TST-7 to TST-13 pass
P2	2 to 3	Reports, 7 more languages, assets and bundles, customer price agreements, reminders, offline edits, presence, margin view, more starter templates, load test	TST-6 passes; v1.1 released
P3	3 to 5	Hosted edition: directory, sign-up, Stripe, plan limits, custom domains, managed email, EU data location, support access, staff console, legal pages, outside security review	Security review findings closed; first 3 hosted pilot workspaces live

PHASE	WEEKS	SCOPE	EXIT TEST
P4	2 to 3	API, OpenAPI, webhooks, HubSpot, Salesforce, Slack and Teams alerts, instant-estimate widget	Integration tests pass against HubSpot and Salesforce developer accounts
P5	1 to 2	Pay on accept, draft from a pasted request	Test-mode payments and suggested-field flow pass end to end

Total: 15 to 23 weeks. Self-deployed v1.0 ships at the end of P1, in week 7 to 10.

14 Success measures

MEASURE	TARGET
Deploy time, click to first saved quote	Under 10 minutes for 9 of 10 first-time testers
Prototype parity	Every fixture matches or is documented
Pricing correctness after launch	Zero reported total errors that the trace can't explain
Free-plan fit	A 50-person team stays under 50% of every daily limit
Customer page speed	Under 1.5 seconds on a mid-range phone on 4G
Hosted pilots	3 pilot workspaces sending real quotes by the end of P3

The self-deployed edition has no telemetry, so its usage isn't measured. GitHub stars, forks, and issues are the only public signal.

15 Risks

RISK	EFFECT	RESPONSE
The button needs a GitHub or GitLab account	Non-technical buyers stall at the GitHub step	Hosted edition for that buyer; a screenshot guide for the button path
Cloudflare changes free limits or button behavior	Self-deployed copies hit limits or new deploys fail	Usage page and warnings; adapters keep a move to Workers Paid a setting, not a rebuild; TST-7 every release
Update workflow can't trigger a build in the customer's copy	Copies fall behind	P0 tests both paths in UPD-3
Template authoring is too hard for customers	Teams never get past setup	Blocks (PRC-10), starter templates, setup service
Public source invites copies	Someone hosts it as a competing service	License (D-1) forbids offering it as a hosted service
Email from the free edition lands in spam	Customers miss quotes	Default send opens the rep's own email app (EML-1)
E-signature rules differ by country	Some acceptances may not hold up	Lawyer review before hosted launch; audit trail on every acceptance
Prototype fixtures leak client data	Breach of trust with the source client	Fixtures stripped to numbers and structure, kept in a private repo, never in the public one

RISK	EFFECT	RESPONSE
Durable Object single writer per workspace	Very large teams queue	Soft limit is 1,000 requests a second per object, far above any quoting team

16 Defaults set in this version

ID	DEFAULT
D-1	License: Elastic License 2.0. Free to use, change, and self-deploy; not allowed to offer as a hosted service
D-2	Repo: `github.com/abchiaravalle/amplifi-quotes`, private through P0 and P1, public at v1.0
D-3	Product name: amplifi Quotes, pending a trademark check
D-4	Self-deployed copies get every feature; hosted plans charge for hosting, domains, email, backups, and support
D-5	Hosted plans and prices as in section 10
D-6	The hosted edition runs under ACWebDev, LLC on amplifi's Cloudflare account on Workers Paid
D-7	Workspace data lives in one SQLite Durable Object per workspace; D1 only for the hosted directory
D-8	Sign-in uses passkeys and email links, with optional Google or Microsoft sign-in. No passwords
D-9	Prototype parity fixtures stay private
D-10	The P0 deploy test runs on a brand-new Cloudflare account and GitHub account

Appendix A Permissions

O = Owner, A = Admin, P = Pricing editor, Ap = Approver, S = Sales, V = Viewer, C = Customer (through the quote link). "own" means quotes the member owns; "team" means all quotes when the workspace setting "Sales see team quotes" is on.

ACTION	O	A	P	AP	S	V	C
Create quote	yes	yes	yes	yes	yes	no	no
Edit draft	yes	yes	own	yes	own	no	no
View quotes	all	all	all	all	own, team	all	linked revision only
Send quote	yes	yes	own	yes	own	no	no
Revise sent quote	yes	yes	own	yes	own	no	no
Revise accepted quote (change order)	yes	yes	no	no	no	no	no
Withdraw quote	yes	yes	own	yes	own	no	no
Approve or reject	yes	yes	no	yes, not own	no	no	no
Override a price or rate on a quote	yes	yes	yes	yes	if allowed	no	no
See cost and margin	yes	yes	yes	if allowed	if allowed	if allowed	no
Accept, decline, ask a question	no	no	no	no	no	no	yes
Edit catalog, rates, regions, tax	yes	yes	yes	no	no	no	no
Edit and publish templates	yes	yes	yes	no	no	no	no
Accept a changed pricing example	yes	yes	yes	no	no	no	no
Manage customers and contacts	yes	yes	yes	yes	yes	no	no
Manage members and roles	yes	yes	no	no	no	no	no
Branding, entities, email templates	yes	yes	no	no	no	no	no
Integrations, API keys, webhooks	yes	yes	no	no	no	no	no
View reports	yes	yes	yes	yes	own	yes	no
View audit log	yes	yes	no	no	no	no	no
Export workspace	yes	no	no	no	no	no	no
Restore backup	yes	no	no	no	no	no	no
Billing (hosted)	yes	no	no	no	no	no	no
Grant support access (hosted)	yes	no	no	no	no	no	no
Delete workspace	yes	no	no	no	no	no	no

Appendix B Formula grammar

```
expr       := or
or         := and ( "or" and )*
and        := not ( "and" not )*
not        := "not" not | compare
compare    := sum ( ( "=" | "!=" | "<" | "<=" | ">" | ">=" ) sum )?
sum        := product ( ( "+" | "-" ) product )*
product    := unary ( ( "*" | "/" ) unary )*
unary      := "-" unary | primary
primary    := number | string | "true" | "false"
            | ref | call | "(" expr ")"
ref        := name ( "." name )*
call       := name "(" ( expr ( "," expr )* )? ")"
number     := digits ( "." digits )?
string     := '"' chars '"'
name       := letter ( letter | digit | "_" )*
```

References. `input.<id>`, bare input ids, `values.<id>`, `rate.<name>`, `item.<attribute>`, `entity.<attribute>`, `customer.<field>`, `site.<field>`, `option.<parameter>`, `quote.<field>` (for example `quote.valid_from`), `today`.

Functions. `if(cond, a, b)`, `min(a, b, ...)`, `max(a, b, ...)`, `round(x, step)`, `ceil(x)`, `floor(x)`, `abs(x)`, `clamp(x, lo, hi)`, `sum(group, expr)`, `count(group, cond)`, `lookup(table, key)`, `coalesce(a, b, ...)`, `days_between(d1, d2)`, `workdays_between(d1, d2)`.

Limits. 500 nodes per formula, no recursion, no user-defined functions, no loops outside `sum` and `count` over a repeating group. Division by zero is a publish-time error when the divisor is a constant and a runtime line error otherwise, shown in the builder and blocking send.

Appendix C Quote status moves

FROM	TO	WHO OR WHAT
Draft	Pending approval	Rep submits and an approval rule fires
Draft	Sent	Rep sends and no rule fires
Pending approval	Approved	Approver
Pending approval	Rejected	Approver
Approved	Sent	Rep
Approved	Draft	Any edit (APR-5)
Rejected	Draft	Rep edits
Sent	Viewed	Customer opens the link
Sent or Viewed	Accepted	Customer accepts
Sent or Viewed	Declined	Customer declines
Sent or Viewed	Expired	Expiry alarm
Sent or Viewed	Withdrawn	Rep or Admin
Sent, Viewed, Declined, or Expired	Draft (new revision)	Rep revises
Accepted	Draft (change-order revision)	Admin (QT-5)

Every move writes a `quote_events` row and an audit entry in the same transaction.

Appendix D Sources

Checked October 20, 2026.

FACT	SOURCE
Workers free plan: 100,000 requests a day, 10 ms CPU, 50 subrequests; Paid $5 a month	developers.cloudflare.com/workers/platform/limits, /workers/platform/pricing
Static asset requests are free and unlimited	developers.cloudflare.com/workers/static-assets/billing-and-limitations
Deploy to Cloudflare button behavior and limits	developers.cloudflare.com/workers/platform/deploy-buttons
Workers Builds: 3,000 minutes a month free, 1 concurrent, 20-minute timeout	developers.cloudflare.com/workers/ci-cd/builds/limits-and-pricing
Durable Objects free plan: SQLite only; 100,000 requests and 13,000 GB-seconds a day; 5 million rows read, 100,000 written; 5 GB	developers.cloudflare.com/durable-objects/platform/pricing, /durable-objects/platform/limits
Durable Object point-in-time restore, 30 days; transactions through `transactionSync`	developers.cloudflare.com/durable-objects/api/sqlite-storage-api
WebSocket hibernation: no duration charge while hibernating	developers.cloudflare.com/durable-objects/best-practices/websockets
Durable Objects EU jurisdiction	developers.cloudflare.com/durable-objects/reference/data-location
D1 free plan: 10 databases, 500 MB each, 100,000 rows written a day	developers.cloudflare.com/d1/platform/limits, /d1/platform/pricing
Email Service: arbitrary recipients need Workers Paid; 3,000 a month included	developers.cloudflare.com/email-service/platform/pricing
Browser Rendering free plan: 10 minutes a day	developers.cloudflare.com/browser-rendering/platform/pricing
Workers block `eval` and `new Function`	developers.cloudflare.com/workers/runtime-apis/web-standards
Rate Limiting binding counts per location and can lag	developers.cloudflare.com/workers/runtime-apis/bindings/rate-limit
Workers Logs free: 200,000 a day, 3-day retention	developers.cloudflare.com/workers/observability/logs/workers-logs
Cron Triggers: 5 per account on free	developers.cloudflare.com/workers/platform/limits
Cloudflare for SaaS: 100 hostnames included, $0.10 each after	developers.cloudflare.com/cloudflare-for-platforms/cloudflare-for-saas/plans
Turnstile free: 20 widgets, 10 hostnames each	developers.cloudflare.com/turnstile/plans
R2 needs a payment method even within the free allowance	community.cloudflare.com/t/why-using-r2-free-tier-involves-giving-card-info/945179
Cloudflare Access free up to 50 users	community.cloudflare.com/t/what-happens-if-i-exceed-50-users/479340
Resend free: 3,000 a month, 100 a day	resend.com/docs/knowledge-base/account-quotas-and-limits
SMTP2GO free: 1,000 a month, 200 a day	support.smtp2go.com/hc/en-gb/articles/223087947-Free-Plan
SimpleWebAuthn runs on Cloudflare Workers	jsr.io/@simplewebauthn
Drizzle durable-sqlite driver	orm.drizzle.team/docs/get-started/do-new

FACT	SOURCE
Pushes made with `GITHUB_TOKEN` don't start new workflow runs	docs.github.com/en/actions/concepts/security/github_token
Password check cost: bcrypt cost 10 at 57 ms (native) and 70 ms (pure JS); passkey signature check 0.08 ms	Measured on an Apple M1 Max, October 20, 2026